

Acceptance Test Plan – MusicCoursePlatform

1. Introduction

1.1 Purpose

This document defines the acceptance testing plan for the MusicCoursePlatform system. The goal is to ensure that the system satisfies all functional and non-functional requirements before deployment.

1.2 Scope

The test plan covers:

- User authentication
- Profile management
- Time slot management
- Lesson booking
- Dashboard functionality
- System usability and performance

2. Acceptance Criteria

The system will be accepted if all of the following measurable criteria are satisfied:

Functional Criteria:

- All core functionalities work as expected with no critical defects
- Users can successfully register, log in, manage profiles, create/edit/delete time slots, book and cancel lessons, and view dashboard data
- No data loss occurs on page refresh or application restart
- Duplicate bookings are prevented by the system
- Form validation rejects invalid inputs and displays actionable error messages

Usability Criteria:

- A new user can locate and complete a booking without confusion
- All UI labels, buttons, and error messages are clear without needing external guidance
- Field-level validation messages appear for relevant input fields

Performance Criteria:

- Dashboard loads within 3 seconds with many booking records
- Booking confirmation is displayed within 1 second of action
- System remains stable and consistent after multiple sequential booking operations

3. Coverage Matrix

Requirement / User Story	Test ID(s) Covered
User registration and login	FT01, FT02, FT03, FT04
Teacher profile management	FT05, FT06
Manage time slots	FT07, FT09, FT10
View available time slots	FT10
Book and cancel lessons	FT12, FT13, FT14
Dashboard view	FT15
Input validation	FT16
System consistency	FT17
Ease of navigation	UT01
Language / UI clarity	UT02
Form validation usability	UT03
System performance under load	PT01, PT03
Fast UI updates	PT02

4. Acceptance Test Design

4.1 Functional Tests

Test ID	Feature	Test Description	Expected Result	Actual Result	Status
FT01	Authentication	Register a new user	Account created and stored in DB	UserDAOTest & UserServiceTest confirm user creation and DB insertion work correctly	Pass
FT02	Authentication	Login with valid credentials	User redirected to dashboard	UserServiceTest confirms login with valid credentials returns correct user	Pass

FT03	Authentication	Login with invalid credentials	Error message displayed	LoginControllerTest ran but <code>languageCombo</code> was null during setup — controller-level UI assertion not fully verified	Partial
FT04	Authentication	Logout	User redirected to login page	SessionManagerTest (14 tests) fully passed — session clearing confirmed	Pass
FT05	Profile Management	Create teacher profile	Profile saved successfully	TeacherProfileDAOTest & TeacherServiceTest confirm profile creation and persistence	Pass
FT06	Profile Management	Edit teacher profile	Changes updated immediately	TeacherProfileDAOTest & TeacherServiceTest confirm update queries execute and reflect changes	Pass
FT07	Time Slot Mgmt	Create time slot	Slot saved and visible	TimeSlotDAOTest & TimeSlotServiceTest confirm slot creation and retrieval	Pass
FT08	Time Slot Mgmt	Edit time slot	Changes reflected immediately	TimeSlotDAOTest & TimeSlotServiceTest confirm update queries execute correctly	Pass
FT09	Time Slot Mgmt	Delete time slot	Slot removed from system	TimeSlotDAOTest & TimeSlotServiceTest confirm deletion and verify slot no longer retrievable	Pass
FT10	Lesson Booking	View available slots	Slots displayed correctly	TimeSlotServiceTest confirms available slot queries return correct results	Pass
FT11	Lesson Booking	Book a lesson	Booking saved and slot unavailable	BookingServiceTest confirms booking is created and slot status updated to BOOKED	Pass
FT12	Lesson Booking	Book already reserved slot	Booking prevented	BookingServiceTest confirms system checks slot status before allowing booking	Pass
FT13	Lesson Booking	Cancel booking	Slot becomes available again	BookingServiceTest confirms cancellation updates booking status and releases slot	Pass
FT14	Dashboard	View scheduled lessons	Correct data displayed	BookingViewControllerTest (4 tests) passed — learner bookings retrieved correctly	Pass

FT15	Validation	Submit invalid inputs	Error messages shown	LoginControllerTest partially executed; languageCombonull prevented full UI validation check	Partial
FT16	Consistency	Refresh after booking	Data remains consistent	BookingDAOTest & BookingServiceTest confirm data persistence across operations	Pass
FT17	Consistency	System consistency across roles	Data consistent for all user types	ModelTest (20 tests) fully passed — all model data structures verified	Pass

4.2 Usability Tests

Test ID	Scenario	Test Description	Expected Outcome	Notes
UT01	Navigation	Can user find booking feature easily?	User completes task without confusion	Not covered by automated tests — requires manual observation session
UT02	Interface Clarity	Are buttons and labels understandable?	User understands interface intuitively	Not covered by automated tests — requires manual review of UI labels
UT03	Form Input	Are invalid inputs handled clearly?	Clear and helpful error messages shown	Partially tested via LoginControllerTest and SignupControllerTest; full UI verification requires manual testing due to JavaFX setup limitations

4.3 Performance & Reliability Tests

Test ID	Scenario	Test Description	Expected Outcome	Notes
PT01	System Load	Load dashboard with many bookings	Loads within 3 seconds	Not covered by automated unit tests — requires manual testing with populated database
PT02	UI Response	Perform booking action	Response within 1 second	Not covered by automated unit tests — requires manual timing during execution
PT03	Bulk Operations	Multiple bookings in sequence	System remains stable	BookingServiceTest executed 18 sequential DB operations without errors, confirming backend stability

4.4 Automated Unit Test Results Summary

The following unit tests were executed using Maven Surefire and JaCoCo on April 13, 2026. All 228 tests passed with BUILD SUCCESS (total execution time: 12.732 seconds).

Test Suite	Tests Run	Passed	Failed	Notes
PasswordUtilTest	22	22	0	All password hashing and validation logic verified
UserDAOTest	16	16	0	All CRUD operations on USERS table verified
BookingDAOTest	9	9	0	All booking DB operations verified
TeacherProfileDAOTest	8	8	0	All teacher profile DB operations verified
TimeSlotDAOTest	9	9	0	All time slot DB operations verified
LearnerProfileDAOTest	6	6	0	All learner profile DB operations verified
TeacherProfileViewControllerTest	4	4	0	One reflection-based assertion (loadSchedule formatter) silently skipped
LoginControllerTest	7	7	0	languageCombo null in 5 of 7 setups — assertions caught internally, not truly verified
SessionManagerTest	14	14	0	Full session lifecycle verified
BookingViewControllerTest	4	4	0	Booking retrieval for learner profile verified
SignupControllerTest	9	9	0	All 8 student/teacher signup scenarios caught null pointer — assertions internally skipped
TeacherDashboardControllerTest	16	16	0	All teacher dashboard logic verified
StudentDashboardControllerTest	11	11	0	All 6 booking/slot scenarios caught null localizationManager logic verified
ModelTest	20	20	0	All model classes and field logic verified
UserServiceTest	23	23	0	Full registration, login, duplicate detection verified
TimeSlotServiceTest	17	17	0	Full slot lifecycle verified
TeacherServiceTest	15	15	0	Full teacher profile service logic verified

BookingServiceTest	18	18	0	Full booking lifecycle including cancellation verified
TOTAL	228	228	0	BUILD SUCCESS

5. Tools and Environment

- Testing Type: Automated Unit Testing + Manual Testing
- Platform: JavaFX Desktop Application
- Backend: Java + Database (MariaDB/MySQL)
- Tools: Maven, JDK 17+
- OS: Windows / macOS / Linux

6. Test Execution Plan

- Tests will be performed manually
- Each test will be marked:
 - Pass (expected result achieved)
 - Fail (issue found)
- Issues will be fixed before final acceptance

7. Acceptance Decision

Based on the automated test run executed on April 13, 2026, the MusicCoursePlatform is conditionally ready for acceptance, pending completion of manual testing phases.

Automated Testing: PASSED — 228/228 unit and integration tests passed with zero failures.

Pending Manual Verification:

- Usability tests UT01, UT02, and UT03 require observation sessions with representative users
- Performance tests PT01 and PT02 require timed manual execution with a populated database
- UI-level behaviour for FT03 and FT15 (invalid login / form validation) requires manual verification — LoginControllerTest assertions were not fully executed due to JavaFX languageCombo initialisation limitations in the test environment.

8. Conclusion

This Acceptance Test Plan ensures that the MusicCoursePlatform system is thoroughly validated against its requirements. Successful completion confirms readiness for deployment.